

Test 1 solutions and grading schemes

Please let me know about any typos/errors.

- Every solution MUST be written using complete sentences. Include all necessary reasoning in a correct order so as to construct a complete and coherent logical argument.
- You may use any definition/theorem we did in class (clearly specify what result you are using), unless the question asks you to prove something standard. Ask me in case of any doubt.
- Upload your answers on moodle. Separate pdf/jpeg file for each problem.

For any discussion of grading do the following. (0) First understand a correct solution and the grading scheme. (1) Then understand what you have written and be ready to explain the content (including any mistakes) fully and lucidly. For unclear writing, beyond a reasonable limit, it is not the job of the grader to decipher it and search for what might be valid elements. (2) Then have a discussion with the respective grader (see below) and ask them to inform me of any possible change in your score. (3) If you are still not satisfied, talk to me.

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0. [12 points] Let $f : V \rightarrow W$ be a linear map between finite dimensional vector spaces.

(a) [2 points] Show that the image $f(V)$ is a subspace of W .

Standard. Nonempty + closed under addition and scalar multiplication by linearity equations.

(b) [5 points] Let $S = \{z_1, \dots, z_\ell\} \subset V$. For each statement below, state if it is true (and prove it) or false (and give a simple counterexample).

1. [1] If S is linearly independent, then $f(S)$ is linearly independent. **False.**

Take $f =$ the zero map.

2. [1.5] If $f(S)$ is linearly independent, then S is linearly independent. **True.**

Let $\sum c_i z_i = 0$. Apply f . By linearity $\sum c_i f(z_i) = 0$. By independence of $f(S)$ each c_i is 0.

3. [1.5] If S spans V , then $f(S)$ spans $f(V)$. **True.**

For any $z \in V$ we have $z = \sum c_i z_i$ for some scalars c_i . Apply f . By linearity $f(z) = \sum c_i f(z_i)$.

4. [1] If $f(S)$ spans $f(V)$ then S spans V . **False.**

Take $f =$ the zero map and S any set not spanning V , e.g. $S = \{0\}$ with V nonzero.

(c) [5 points] Complete the following (slightly different) proof of the rank-nullity theorem for f .

Let $w_1 = f(v_1), \dots, w_r = f(v_r)$ be a basis for $f(V)$, i.e., a basis for the image of f .

Claim: [1] The vectors $v_1, \dots, v_r$, together with a basis $x_1, \dots, x_n$ of the kernel (or null space) of f , form a basis of V .

Proof of spanning: [2] Consider an arbitrary vector y in V . Then $f(y) = c_1 f(v_1) + \dots + c_r f(v_r)$. Consider $z = y - (c_1 v_1 + \dots + c_r v_r)$. By linearity $f(z) = f(y) - f(c_1 v_1 + \dots + c_r v_r) = 0$. So z is in the kernel of f and is therefore a linear combination of $x_1, \dots, x_n$. Hence $y = z + c_1 v_1 + \dots + c_r v_r$ is a linear combination of $v_1, \dots, v_r$ and $x_1, \dots, x_n$.

Proof of independence: [2] Suppose there is a linear relation $a_1v_1 + \dots + a_rv_r + b_1x_1 + \dots + b_nx_n = 0$. Then applying f and using linearity gives $a_1f(v_1) + \dots + a_rf(v_r) = 0$ because each x_i is in the kernel of f . As $w_1 = f(v_1), \dots, w_r = f(v_r)$ are linearly independent, each $a_i = 0$. Then the supposed relation becomes $b_1x_1 + \dots + b_nx_n = 0$. As $x_1, \dots, x_n$ are linearly independent, each $b_i = 0$ as well.

1. [8 points] Consider the linear map $f : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ given $f(x) = Mx$ where M is a 4×3 matrix. M is not given but you are told that doing a series of column operations $op_1, op_2, \dots, op_7$ to M gives

the matrix $N = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 3 & 4 \\ 20 & 30 & 40 \end{bmatrix}$. Note that one can recover M from N using the operations.

Give an ordered basis $\mathbf{B}$ of $\mathbb{R}^3$ and an ordered basis $\mathbf{C}$ of $\mathbb{R}^4$ such that each entry of the matrix of f with respect to these new bases is 0 or 1, with the fewest 1's possible. **Remark:** the minimum number of nonzero entries is exactly the rank of the matrix. This needs proof!

Your answer must be of the following form. For each basis, if possible, give a list of vectors. If the given information is not enough to write actual vectors, then give an algorithmic procedure in terms of the given information. Justify.

[See board image for a solution and the grading scheme: 1+2+1+2+2 for steps 0, 1, 2, 3, 4.]

2. [10 points] Let $T : U \rightarrow V$ and $S : V \rightarrow W$ be linear maps between finite dimensional vector spaces.

Prove the four inequalities below. Here r stands for the rank (i.e., dimension of the image) of a linear map and n stands for its nullity (i.e., dimension of the kernel). Possibly useful considerations: $T(U)$ is a subspace of V . By restricting S to this subspace, we have the linear map $S|_{T(U)} : T(U) \rightarrow W$.

(a) $r(ST) \leq r(S)$. (b) $n(T) \leq n(ST)$. (c) $r(ST) \leq r(T)$. (d) $n(ST) \leq n(S) + n(T)$.

[See board image for a solution. Grading scheme: 1.5 for (a), 1.5 for (b), 2 for (c), 5 for (d).]

3. [5 + 5 = 10 points, credit based on quality of reasoning.] (a) For v in $\mathbb{R}^2$, define $F(v)$ to be the reflection of v in the line $y = 5x$. Find the matrix of the linear operator F with respect to the standard basis. Then find a basis w_1, w_2 of $\mathbb{R}^2$ with respect to which the matrix of F is diagonal.

It is simpler to do the second task first and then change basis. Take w_1 along the line $y = 5x$, e.g. $w_1 = [1, 5]^t$, and take w_2 perpendicular to the line $y = 5x$, e.g. $w_2 = [-5, 1]^t$. Then $F(w_1) = w_1$ and $F(w_2) = -w_2$, so the matrix of F with respect to the ordered basis w_1, w_2 is the diagonal matrix D with entries 1, -1.

Therefore the matrix of F with respect to the standard basis is $P^{-1}DP$ where P is the change of basis matrix from w_1, w_2 to the standard basis. So $P^{-1} = \begin{bmatrix} w_1 & w_2 \end{bmatrix}$, $P = \begin{bmatrix} w_1 & w_2 \end{bmatrix}^{-1}$ and the matrix of F with respect to the standard basis is

$$P^{-1}DP = \begin{bmatrix} 1 & -5 \\ 5 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \frac{1}{26} \begin{bmatrix} 1 & 5 \\ -5 & 1 \end{bmatrix} = \frac{1}{13} \begin{bmatrix} -12 & 5 \\ 5 & 12 \end{bmatrix}.$$

(b) Show that any 2×2 matrix A satisfying $A^2 =$ the identity matrix I is either $\pm I$ or conjugate to the matrix of a reflection in some line, i.e., of the form $P^{-1}(\text{the matrix of a reflection})P$.

There is more than one possible approach. Here is perhaps the simplest. $A^2 = I$ forces $A(Av) = v$ for any vector v . So if we find v such that v and Av are linearly independent (i.e., such that Av is not a multiple of v) then v, Av is a basis of $\mathbb{R}^2$. With respect to this basis, the matrix of the linear map $f(v) = Av$ is $R = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. So $A = \begin{bmatrix} v & Av \end{bmatrix} (R, \text{i.e., the matrix of reflection in } y = x) \begin{bmatrix} v & Av \end{bmatrix}^{-1}$.

So it remains to examine the case in which, for every v in $\mathbb{R}^2$, the vector Av is a scalar multiple of v . Then this scalar must be the same for every v . (Reason: If $Av = bv$ and $Aw = cw$ with $b \neq c$, then $A(v + w) = bv + cw$ is not a scalar multiple of $v + w$. This was Artin problem 3.3 in chapter 3, done in Tutorial 4.) So A must be a scalar matrix, which then must be $\pm I$ because $A^2 = I$.

Remark: one can revisit this problem using eigenanalysis. $A^2 = I$ shows that the only possible eigenvalues are $1, -1$. If both occur then, with respect to the corresponding eigenbasis, $f(v) = Av$ has a diagonal matrix D with entries $1, -1$. So A is similar to D , which is the matrix of reflection in the X -axis. (Note that f itself need not be a reflection because the 1 -eigenvector and the (-1) -eigenvector need not be perpendicular to each other.)

It remains to examine the case in which the set of eigenvalues of A is a proper subset of $\{-1, 1\}$. The following claims suffice. If $A^2 = I$, then A must have at least one eigenvalue (which then must be 1 or -1). Additionally, if only one of $1, -1$ is an eigenvalue of A , then $A = \pm I$. Proof?

4. Short independent questions Note the crossed out and corrected typo in 4(C). For this reason I took the best two out of the four parts in this question instead of the planned three.

(A) [3 points] For an $r \times c$ matrix A , consider the functions $f(x) = Ax$ and $g(y) = A^T y$. Suppose g is a surjective function. What does that imply about the function f ? Give the strongest answer that you can. Is the converse of your implication true?

The following are equivalent by our table. So g is surjective if and only if f is injective.

- 1) $g(y) = A^T y$ is surjective.
- 2) $A^T y = b$ has a solution y for every b in $\mathbb{R}^c$.
- 3) $RREF(A^T)$ has a pivot in each row.
- 4) All c rows of A^T are linearly independent.
- 5) All c columns A are linearly independent.
- 6) $Ax = 0$ has only the trivial solution.
- 7) $f(x) = Ax$ is injective.

As $\text{ranks}(A) = \text{rank}(A^T)$, one can also say at steps 4 and 5 that A^T and A have full rank c .

Remark: It is sloppy and incorrect to say “ A^T has a pivot in each row so A has a pivot in each column.” The statements about pivots are true for RREF, not the original matrix. And doing row operations to a matrix corresponds to doing *column* operations on the transpose. A little more care is needed in the reasoning.

(B) [3 points: 1 each] You are given a set of vectors $v_1, \dots, v_k$ in some $\mathbb{R}^n$. State how you will solve the following. Your answer must be completely algorithmic. You may assume as black box ONLY row reduction and its connection with matrix multiplication. You may NOT say “solve this equation” or “invert this matrix”. If that’s what you need to do, you must actually state a programmable procedure, including precise instructions about how you will make any required decision.

1. Are $v_1, \dots, v_k$ linearly independent?
2. Do $v_1, \dots, v_k$ span $\mathbb{R}^n$?
3. Assume $k = n$ and that $v_1, \dots, v_n$ form a basis of $\mathbb{R}^n$. For a given vector b , how will you find the coordinates of b in terms of this new basis?

Make a matrix M whose columns are $v_1, \dots, v_k$.

1. $v_1, \dots, v_k$ are linearly independent iff $Mx = 0$ has only the trivial solution, which happens if and only if $RREF(M)$ has a pivot in each column (no free variables).

2. $v_1, \dots, v_k$ span $\mathbb{R}^n$ iff $Mx = b$ has a solution for each b in $\mathbb{R}^n$, which happens if and only if $RREF(M)$ has a pivot in each row, i.e., has no zero row (otherwise one can create inconsistency in the augmented matrix).

3. Solve $Mx = b$ by row reduction. As columns of M form a basis, row reduction gives I , so a unique solution can be simply read off.

(C) [3 points] Prove multiplicativity of the determinant function as follows. For a fixed $n \times n$ matrix A define the function ~~$f(AX)$~~ $f(X) = \det(AX) = \det(Ax_1, Ax_2, \dots, Ax_n)$ where X is a square matrix with column vectors $x_1, \dots, x_n$. You may use that determinant is the unique function such that ... as a function of the n columns of the matrix, it is multilinear, alternating and takes I to 1.

Linearity in any one slot follows from $A(x + x') = Ax + Ax'$ and multilinearity of the determinant. If $x_i = x_j$, then $Ax_i = Ax_j$, so $f(X) = \det(AX)$ vanishes. If $\det(A) \neq 0$ then the function $f(X)/\det(A)$ of columns of X is multilinear, alternating and takes I to 1, so must equal $\det(X)$. If $\det(A) = 0$ then A is not invertible, so neither is AX and hence $\det(A)\det(X) = 0 = \det(AX)$.

(D) [4 points] Let $A = \begin{bmatrix} a_1 & 0 \\ 0 & a_2 \end{bmatrix}$, $B = \begin{bmatrix} b_1 & 0 & 0 \\ 0 & b_2 & 0 \\ 0 & 0 & b_3 \end{bmatrix}$ and $M_{2 \times 3}$ = the vector space of 2×3 matrices.

For any X in $M_{2 \times 3}$, define $T(X) = AX - XB$. Let M be the matrix of the linear map T with respect to some bases.

- Show that $\det(M) = 0$ if and only if $\{a_1, a_2\} \cap \{b_1, b_2, b_3\}$ is nonempty.
- Calculate $\text{rank}(T)$.

With respect to the standard basis of $M_{2 \times 3}$ consisting of the 6 matrix units E_{pq} , the 6×6 matrix of T is diagonal with diagonal entries $a_p - b_q$. (Calculate and see.) This matrix is invertible precisely when every $a_p - b_q \neq 0$. Its rank is the number of nonzero values among the 6 numbers $a_p - b_q$.